Jonathan Colaço Carr

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Education

Harvard University, Cambridge, MA
Ph.D. in Computer Science. Expected completion: June 2032.
Advisor: Milind Tambe

McGill University/Mila, Montreal, Canada
M.Sc. in Computer Science, August 2026.
Advisors: Prakash Panangaden and Doina Precup
Thesis title: *An analysis of optimization targets for value alignment within the Markov occupancy problem*

McGill University, Montreal, Canada
B.Sc. in Honours Mathematics, May 2023.
Minors in Computer Science and Physics

Honors and Awards

2026 FRQNT Doctoral Scholarship and Supplement, $106,000
Four-year scholarship for Quebec students, determined by academic record and research potential. Includes a $6,000 supplement for ranking in the top 10 of approximately 450 applicants pursuing out-of-province studies.

2024 MITACS Globalink Research Award, $6,000
Research award for Canadian graduate students studying abroad, determined by expected research quality.

2024 McGill Graduate Mobility Award, $8,900
Scholarship for McGill graduate students studying abroad, determined by academic standing.

2023 NSERC Canada Graduate Scholarship – Master's, $17,500
Graduate scholarship determined by academic excellence, research potential, and interpersonal skills.

2022 McGill Computer Science Undergraduate Research Award, $7,000
Research award determined by academic record and research aptitude.

2021 McGill Space Institute Summer Undergraduate Research Award, $7,000
Research award determined by academic record and extracurricular leadership.

2020 RBC x Microsoft AI for Social Impact Challenge, 2nd Place Prize, $2,000
National AI competition prize determined by originality, technical excellence, impact, and feasibility. Placed 2nd out of 167 teams from across Canada.

Preprints and Publications

* denotes co-first authorship.

Choo, D., A. Kangaslahti, J. Colaço Carr, A. van Heerden, C. Johnson, and M. Tambe (2026). *Deployment-Constrained Priority Pool Selection: Incentivized Peer Referral for Health Chatbot Adoption in HIV Outreach*. Under review at the 27th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2027).

Colaço Carr, J. (2026). "An analysis of optimization targets for value alignment within the Markov occupancy problem". MA thesis. McGill University (in press).

Colaço Carr, J., P. Panangaden, D. Precup, and B. Van Roy (2026). *Reinforcement learning with pairwise preferences in long-term decision problems*. In: *International Conference on Machine Learning (ICML)*. acceptance rate: 26.6%.

Colaço Carr, J., Q. Sun, and C. Allen (2025). *Focused skill discovery: Learning to control specific state variables while minimizing side effects*. In: *Reinforcement Learning Journal (RLJ)*. acceptance rate: 39.0%.

Chehbouni*, K., J. Colaço Carr*, Y. More, J. C. Cheung, and G. Farnadi (2025). *Beyond the safety bundle: Auditing the Helpful and Harmless dataset*. In: *Proceedings of the Annual Conference of the Nations of the Americas Chapter of the ACL (NAACL)*. acceptance rate: 22.6%. Selected for an oral presentation.
– An earlier version was selected for a spotlight presentation at the NeurIPS 2024 workshop on Algorithmic Fairness through the Lens of Metrics and Evaluations.

Colaço Carr, J., P. Panangaden, and D. Precup (2024). *Conditions on preference relations that guarantee the existence of optimal policies*. In: *International Conference on Artificial Intelligence and Statistics (AISTATS)*. acceptance rate: 27.6%.

Kennedy, J., J. Colaço Carr, S. Gagnon-Hartman, A. Liu, J. Mirocha, and Y. Cui (2024). *Machine-learning recovery of foreground wedge-removed 21-cm light cones for high-z galaxy mapping*. In: *Monthly Notices of the Royal Astronomical Society (MNRAS)*.

Research Experience

2026–Present **Harvard University**, Cambridge, MA
Student Researcher
Advisor: Milind Tambe
Researching efficient algorithms for AI applications in public health.

2022–2026 **Mila – Quebec AI Institute**, Montreal, Canada
Student Researcher
Advisors: Prakash Panangaden and Doina Precup
Used human models from economics to design sequential decision-making algorithms aligned with human goals. Audited a human preference dataset for label quality, topic coverage, and changes in language-model behaviour after fine-tuning.

2024–2025 **Stanford University**, Stanford, CA
Visiting Student Researcher
Supervisor: Benjamin Van Roy
Extended the maximal lottery social choice rule to sequential decision-making problems and designed game-theory algorithms for computing solutions.

2024 **UC Berkeley (Center for Human-Compatible AI)**, Berkeley, CA
(Summer) Supervisors: Cameron Allen and Stuart Russell
Visiting Student Researcher
Developed a simple method to improve unsupervised reinforcement learning using state abstraction, benefiting both efficiency and safety.

2021 **McGill Space Institute**, Montreal, Canada
Undergraduate Researcher
Supervisor: Adrian Liu
Implemented computer-vision algorithms to recover signals from noisy telescope data.

Work Experience

2024–2025 **Hortus AI**, Berkeley, CA
Lead Software Developer
Led a team of developers (15 in total) to build a consumer reports platform for AI tools. Reported directly to startup founder, Thomas Krendl Gilbert. Designed algorithms to rank and evaluate AI tools in collaboration with the GovAI Coalition. Platform was used by over 100 government officials in its first three months of deployment and secured funding for 5 years.

2025 **The Democratic Engagement Exchange**, Toronto, Canada (remote)
(Fall) *Pilot Researcher*
Conducted a feasibility study on the use of personalized AI assistants for voter education in Canada.

2024 **McGill University**, Montreal, Canada
(Winter) *Teaching Assistant* for COMP 551: Applied Machine Learning
Held weekly office hours and created assignments for a graduate-level course in machine learning.

2022 **AltaML**, Toronto, Canada (remote)
(Winter) *Machine Learning Intern*
Implemented machine learning algorithms to identify cross-selling opportunities.

Extracurricular Experience

2022–2023 **McGill AI Safety x Law Group**, Montreal, Canada
Co-founder
Co-founded and taught an eight-week course on AI safety for McGill Law students.

2019–2020 **RBC x Microsoft AI for Social Impact Challenge**, Montreal, Canada
Team Lead
Led a four-student team to design a textile recycling robot for an AI social impact challenge. Placed 2nd out of 167 teams from across Canada.

2019–2020 **The McGill Tribune**, Montreal, Canada
Editorial Board Member
Collaborated with journalists and graphic designers to publish articles for McGill's student newspaper.

2015 **Youth and Philanthropy Initiative (YPI)**, Montreal, Canada
Team Lead
Led the 1st place team (in a pool of 20 teams) at my high school's YPI competition, earning a $5,000 grant for the Tyndale St-Georges Community Center.